

TECHNICAL COMPETENCIES ADT

	 0	 2.5	 5	 7.5	 10
IL5.1 Develops applications that manage information stored in files , identifying and using connection mechanisms (ADT1)	The project structure is not adequate.	Correct Project Structure.	In addition to all of the above: Performs one of the operations correctly in the file.	In addition to all of the above: Performs all operations correctly in the file.	In addition to all of the above: - All access and closure operations are correct. - The program opens the corresponding file from the path specified in the database.
IL5.2 Develops applications that manage information stored in relational databases , identifying and using connection mechanisms (ADT2) (ADT4)	The necessary components are not installed, configured, or are not functioning correctly.	The necessary components are installed, configured, and functioning correctly.	In addition to all of the above: - Correct Project Structure. - The database is created correctly.	In addition to all of the above: - Use Connectors / Classes correctly to work with relational databases. - The database has data when it is created.	In addition to all of the above: All opening and closing operations are clear and efficient

IL5.3 Manages data persistence by identifying Object-Relational Mapping (ORM) tools and developing applications that utilize them (ADT3)	Improper handling of data persistence.	Proper handling of data persistence.	Correct handling of data and relationship persistence.	All persistence operations are correct.	All persistence operations are correct and efficient (handling of potential user errors).
IL5.5 Program data access components, identifying the characteristics a component must possess and using development tools. (ADT6)	Data access operations are not programmed into a dedicated component; instead, they are distributed across objects in the business or UI layers.	A data access object is programmed to expose only operations that manipulate the data.	In addition to all of the above: • An interface is used for the proper isolation of the data layer. (DAO) • A single implementation of the interface is programmed in a logical and efficient manner	In addition to all of the above: • The interface is correctly designed (method arguments and exception handling)	In addition to all of the above: • Correct implementation of the <i>Singleton design pattern</i>.
IL5.6 Manage exceptions and potential errors by developing applications that properly handle them (ADT1) (ADT2) (ADT4) (ADT5)	Exceptions are not handled.	Only exceptions imposed by the programming language are handled.	In addition to all of the above: • Custom exceptions are created to anticipate potential errors • Exceptions are handled in the method where they are caught.	In addition to all of the above: • Exceptions propagate and are handled correctly in the UI layer.	In addition to all of the above: • Exception handling is comprehensive and efficient.